

Lab 3 - ANOVA Shiny App

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App Description

This Shiny app visualizes how ANOVA results change based on group means and variability. Users can interactively adjust parameters and observe the corresponding changes in boxplots and ANOVA statistics.

App Link

Live App: <https://joohokim.shinyapps.io/lab3/>

Features

Interactive Controls

- **Group Means (3 sliders):** Adjust the mean for each of the three groups
- **Standard Deviation:** Control within-group variability
- **Sample Size:** Change the number of observations per group

Visual Output

- **Boxplots:** Three side-by-side boxplots that update in real-time as sliders are adjusted
- **ANOVA Table:** Displays F-statistic, p-value, degrees of freedom, sum of squares, and mean squares

Key Insight

The app demonstrates how ANOVA detects differences between groups. When group means are far apart (large between-group variation), the F-value increases and p-value decreases, indicating significant differences. When means are similar, the F-value decreases and p-value increases.